

Informe de laboratorio - [Instalacion de Docker en VM Ubuntu 64bits]

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****Entorno:**** Kali Linux X.Y, Linux Mint/Ubuntu 64-bit, VMs (nombres/IPs)

****Objetivo:**** Tener una mini máquina virtual dentro de la VM para optimizar recursos.

Resumen ejecutivo

Una o dos líneas con el resultado principal.

Alcance y limitaciones

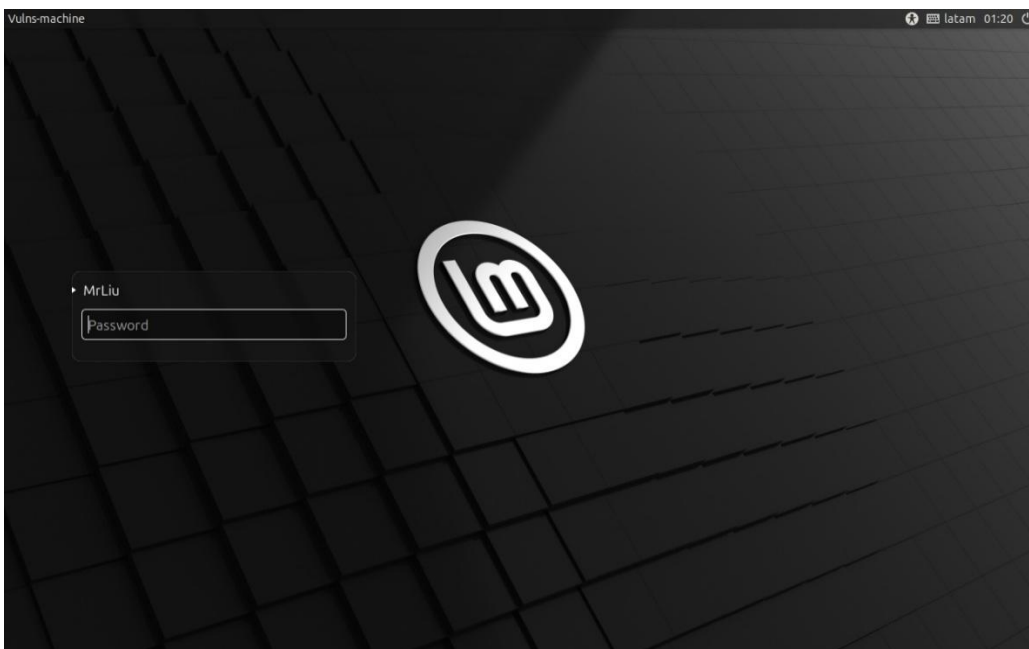
Qué sistemas se probaron; qué no se hizo; notas sobre reconstrucción de pasos previos.

Metodología

Herramientas usadas: `VM`.

Instalacion de Docker en Linux Mint Ubuntu 64 bits

//Inicio de Sesión con contraseña propia.



```
Terminal - mrlui@Vulns-machine: ~
File Edit View Terminal Tabs Help

mrlui@Vulns-machine:~$ sudo apt up
update upgrade
mrlui@Vulns-machine:~$ sudo apt update && sudo apt upgrade -y
[sudo] password for mrlui:
Sorry, try again.
[sudo] password for mrlui:
Sorry, try again.
[sudo] password for mrlui:
Ign:1 http://packages.linuxmint.com zena InRelease
Hit:2 http://packages.linuxmint.com zena Release
Hit:3 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:4 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:7 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
All packages are up to date.
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Calculating upgrade... Done
0 upgraded, 0 newly installed, 0 to remove and 0 not upgraded.
mrlui@Vulns-machine:~$ sudo apt install -y apt-transport-https ca-certificates c
```

// Actualizacion de todos los paquetes del sistema detectados hasta el momento.

```
Terminal - mrlui@Vulns-machine: ~
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apt-transport-https software-properties-common
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.
Need to get 8,966 B of archives.
After this operation, 49.2 kB of additional disk space will be used.
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 apt-transport-https all 2.8.3 [3,970 B]
Get:2 http://packages.linuxmint.com zena/upstream amd64 software-properties-common all 2.3.8 [4,996 B]
Fetched 8,966 B in 0s (35.3 kB/s)
Selecting previously unselected package apt-transport-https.
(Reading database ... 499934 files and directories currently installed.)
Preparing to unpack .../apt-transport-https_2.8.3_all.deb ...
Unpacking apt-transport-https (2.8.3) ...
Selecting previously unselected package software-properties-common.
Preparing to unpack .../software-properties-common_2.3.8_all.deb ...
Unpacking software-properties-common (2.3.8) ...
Setting up apt-transport-https (2.8.3) ...
Setting up software-properties-common (2.3.8) ...
mrlui@Vulns-machine:~$ cu
cupsaccept cupsd cups-genppdupdate cut
cups-browsed cupsdisable cupsreject
cups-calibrate cupsenable cupstestppd
cupsctl cupsfilter curl
mrlui@Vulns-machine:~$ curl -fsSL https://get.docker.com | sh
```

```
Terminal - mrlu@Vulns-machine: ~
File Edit View Terminal Tabs Help

docker-init:
Version:      0.19.0
GitCommit:    de40ad0

=====

To run Docker as a non-privileged user, consider setting up the
Docker daemon in rootless mode for your user:

    dockerd-rootless-setuptool.sh install

Visit https://docs.docker.com/go/rootless/ to learn about rootless mode.

To run the Docker daemon as a fully privileged service, but granting non-root
users access, refer to https://docs.docker.com/go/daemon-access/

WARNING: Access to the remote API on a privileged Docker daemon is equivalent
to root access on the host. Refer to the 'Docker daemon attack surface'
documentation for details: https://docs.docker.com/go/attack-surface/

=====

mrlu@Vulns-machine:~$ sudo systemctl enable docker
```

```
Terminal - mrlu@Vulns-machine: ~
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=====

mrlu@Vulns-machine:~$ sudo systemctl enable docker
Synchronizing state of docker.service with SysV service script with /usr/lib/sys
temd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
mrlu@Vulns-machine:~$ sudo systemctl start docker
mrlu@Vulns-machine:~$
```

// Descargar y Activación de Docker

```
Terminal - mrlu@Vulns-machine: ~
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Docker daemon in rootless mode for your user:

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mrlu@Vulns-machine:~$ sudo systemctl enable docker
Synchronizing state of docker.service with SysV service script with /usr/lib/sys
temd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
mrlu@Vulns-machine:~$ sudo systemctl start docker
mrlu@Vulns-machine:~$ sudo usermod -aG docker $USER
mrlu@Vulns-machine:~$ newgrp docker
mrlu@Vulns-machine:~$
```

// Se Agrega tu usuario al grupo Docker (para no usar sudo)

```
Terminal - mrlu@Vulns-machine: ~
File Edit View Terminal Tabs Help

-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
mrlu@Vulns-machine:~$ sudo systemctl start docker
mrlu@Vulns-machine:~$ sudo usermod -aG docker $USER
mrlu@Vulns-machine:~$ newgrp docker
mrlu@Vulns-machine:~$ docker --version
Docker version 29.4.3, build 055a478
mrlu@Vulns-machine:~$ docker compose version
Docker Compose version v5.1.3
mrlu@Vulns-machine:~$ docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:f9078146db2e05e794366b1bfe584a14ea6317f44027d10ef7dad65279026885
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
1. The Docker client contacted the Docker daemon.
2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
   (amd64)
3. The Docker daemon created a new container from that image which runs the
   executable that produces the output you are currently reading.
4. The Docker daemon streamed that output to the Docker client, which sent it
   to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

mrlu@Vulns-machine:~$
```

//Verificar la instalación y validar la instalación con el contenedor de prueba


```
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File Edit View Terminal Tabs Help

-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable docker
mrlu@Vulns-machine:~$ sudo systemctl start docker
mrlu@Vulns-machine:~$ sudo usermod -aG docker $USER
mrlu@Vulns-machine:~$ newgrp docker
mrlu@Vulns-machine:~$ docker --version
Docker version 29.4.3, build 055a478
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mrlu@Vulns-machine:~$ docker run -d -p 8080:80 --name test-nginx nginx
```

// Validación adicional: levantar un Nginx y probar en navegador

```
Terminal - mrlu@Vulns-machine: ~
File Edit View Terminal Tabs Help

3. The Docker daemon created a new container from that image which runs the
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https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

mrlu@Vulns-machine:~$ docker run -d -p 8080:80 --name test-nginx nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
8e9b53b630de: Pull complete
6e9f32fddb61: Pull complete
382a36159731: Pull complete
57fb71246055: Pull complete
38a7e44929f3: Pull complete
83742381311a: Pull complete
20d41cd6829d: Pull complete
94a543ead0f2: Download complete
46ca967efb5: Download complete

Digest: sha256:a7a9f7e9f549699206cd61c9d14c5434c7ed85e510e837211e0c2d24c9bdbb9ac
Status: Downloaded newer image for nginx:latest
5e181631e5899acbd94a3755993e3662ee3e667b593166a9c0699efb5d65985f
mrlu@Vulns-machine:~$
```

Welcome to nginx! — Mozilla Firefox

Problem loading p x Welcome to Firefo x Welcome to nginx! x + v

← → ↻ http://localhost:8080 ☆ 🔍 ☰

Welcome to nginx!

If you see this page, nginx is successfully installed and working. Further configuration is required for the web server, reverse proxy, API gateway, load balancer, content cache, or other features.

For online documentation and support please refer to nginx.org.
To engage with the community please visit community.nginx.org.
For enterprise grade support, professional services, additional security features and capabilities please refer to f5.com/nginx.

Thank you for using nginx.

```
Terminal - mrlu@Vulns-machine: ~
File Edit View Terminal Tabs Help

mrlu@Vulns-machine:~$ docker run -d -p 8080:80 --name test-nginx nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
8e9b53b630de: Pull complete

6e9f32fdbd61: Pull complete
382a36159731: Pull complete
57fb71246055: Pull complete
38a7e44929f3: Pull complete
83742381311a: Pull complete
20d41cd6829d: Pull complete
94a543ead0f2: Download complete
46ca9676efb5: Download complete

Digest: sha256:a7a9f7e9f549699206cd61c9d14c5434c7ed85e510e837211e0c2d24c9bdb9ac
Status: Downloaded newer image for nginx:latest
5e181631e5899acb94a3755993e3662ee3e667b593166a9c0699efb5d65985f
mrlu@Vulns-machine:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS
PORTS
5e181631e589   nginx     "/docker-entrypoint..." 7 minutes ago  Up 7 minutes
0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  test-nginx
mrlu@Vulns-machine:~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS
PORTS
5e181631e589   nginx     "/docker-entrypoint..." 9 minutes ago  Up 9 minut
es
0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  test-nginx
76c842cde729   hello-world "/hello"                14 minutes ago  Exited (0)
14 minutes ago                                elated_hypatia
mrlu@Vulns-machine:~$
```

// Gestión de contenedores individuales y listar todos los contenedores (activos e inactivos)

```
Terminal - mrlu@Vulns-machine: ~
File Edit View Terminal Tabs Help

94a543ead0f2: Download complete
46ca9676efb5: Download complete

Digest: sha256:a7a9f7e9f549699206cd61c9d14c5434c7ed85e510e837211e0c2d24c9bdb9ac
Status: Downloaded newer image for nginx:latest
5e181631e5899acb94a3755993e3662ee3e667b593166a9c0699efb5d65985f
mrlu@Vulns-machine:~$ docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS
PORTS
5e181631e589   nginx     "/docker-entrypoint..." 7 minutes ago  Up 7 minutes
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mrlu@Vulns-machine:~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS
PORTS
5e181631e589   nginx     "/docker-entrypoint..." 9 minutes ago  Up 9 minut
es
0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  test-nginx
76c842cde729   hello-world "/hello"                14 minutes ago  Exited (0)
14 minutes ago                                elated_hypatia
mrlu@Vulns-machine:~$ docker stop test-nginx
test-nginx
mrlu@Vulns-machine:~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS
PORTS
5e181631e589   nginx     "/docker-entrypoint..." 13 minutes ago  Exited (0)
12 seconds ago                                test-nginx
76c842cde729   hello-world "/hello"                19 minutes ago  Exited (0)
19 minutes ago                                elated_hypatia
mrlu@Vulns-machine:~$ docker start test-nginx
test-nginx
mrlu@Vulns-machine:~$ docker ps -a
CONTAINER ID   IMAGE     COMMAND                  CREATED        STATUS
PORTS
5e181631e589   nginx     "/docker-entrypoint..." 14 minutes ago  Up 5 secon
ds
0.0.0.0:8080->80/tcp, [::]:8080->80/tcp  test-nginx
76c842cde729   hello-world "/hello"                19 minutes ago  Exited (0)
19 minutes ago                                elated_hypatia
mrlu@Vulns-machine:~$
```

// Detener un contenedor específico (ejemplo: test-nginx) e Iniciar un contenedor detenido

Resumen de comandos utilizados:

```
Terminal - mrlu@Vulns-machine: ~
File Edit View Terminal Tabs Help

CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS
5e181631e589   nginx     "/docker-entrypoint..." 9 minutes ago  Up 9 minut
es           0.0.0.0:8080->80/tcp, [::]:8080->80/tcp test-nginx
76c842cde729   hello-world "/hello"                14 minutes ago Exited (0)
14 minutes ago                                elated_hypatia

mrlu@Vulns-machine:~$ docker stop test-nginx
test-nginx
mrlu@Vulns-machine:~$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS
5e181631e589   nginx     "/docker-entrypoint..." 13 minutes ago Exited (0)
12 seconds ago test-nginx
76c842cde729   hello-world "/hello"                19 minutes ago Exited (0)
19 minutes ago                                elated_hypatia

mrlu@Vulns-machine:~$ docker start test-nginx
test-nginx
mrlu@Vulns-machine:~$ docker ps -a
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS
5e181631e589   nginx     "/docker-entrypoint..." 14 minutes ago Up 5 secon
ds           0.0.0.0:8080->80/tcp, [::]:8080->80/tcp test-nginx
76c842cde729   hello-world "/hello"                19 minutes ago Exited (0)
19 minutes ago                                elated_hypatia

mrlu@Vulns-machine:~$ history
 1 sudo apt update
 2 sudo apt upgrade
 3 docker --version
 4 docker compose version
 5 docker run hello-world
 6 docker run -d -p 8080:80 --name test-nginx nginx
 7 docker ps
 8 docker ps -a
 9 docker stop test-nginx
10 docker ps -a
11 docker start test-nginx
12 docker ps -a
13 history
mrlu@Vulns-machine:~$
```

02:41
2026-05-09